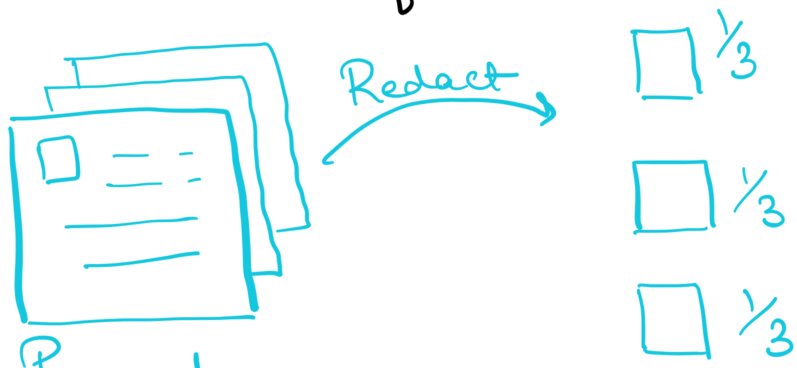
Map Reduce

* Being able to process massive amount of data



Map Reduce

Processing Data

Batch

→ Given data upfront
eg People financial data, Word Count
Say you want to count no of times Harry word comes in Harry Potter books quickly
Another eg is running batch job in intervals like every week/month

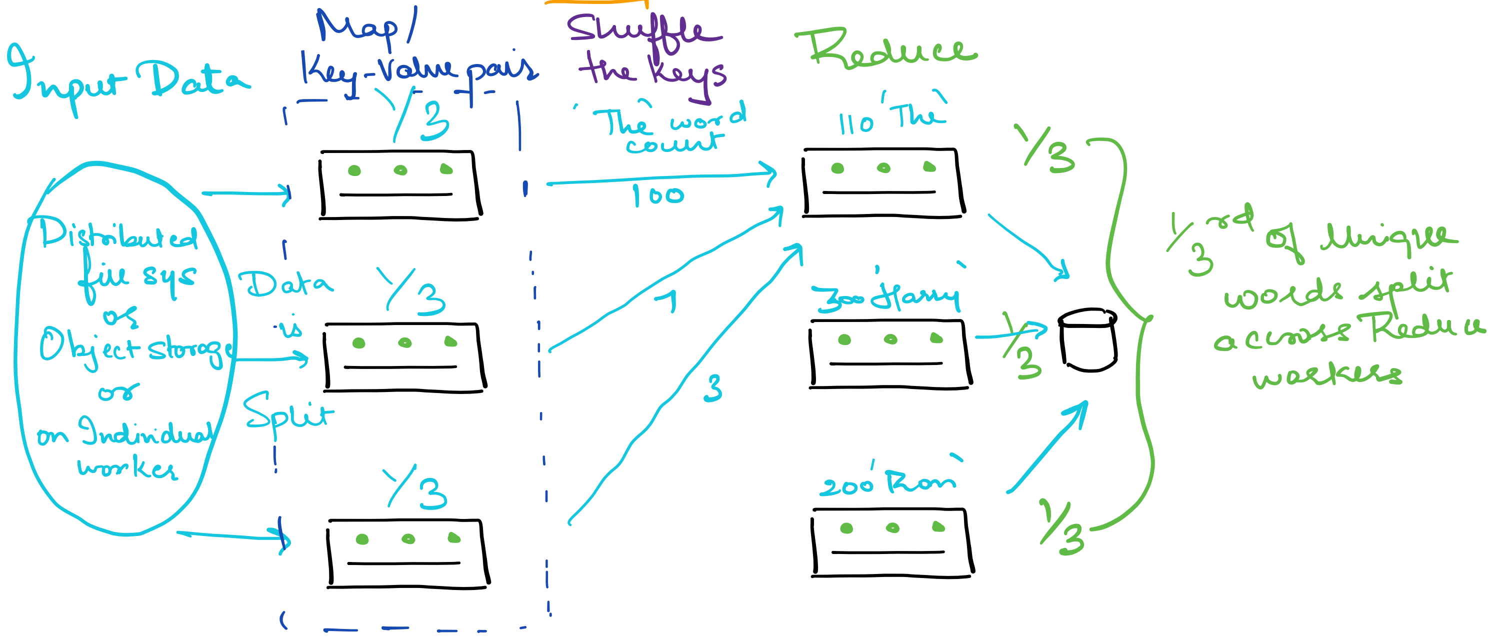
→ Micro Batch processing that executes Batch job quickly like in 30sec imitate Streaming (kind of RT)

Streaming

→ Data is not upfront
eg:- Real time data like a credit card transaction that happens in RT and we want to Redact some info in RT

→ In our context, as Streaming takes RT data it can have an input source like a Message Queue

Map Reduce



Workers/Servers

→ Each worker has some amount of data and the co-ordination done through Master/Leader process

→ In harry potter example say the 1st 3 set of workers will each get 1/3rd of the Pages. So Map step is basically counting the occurence of every single word by each of those 3 workers. Map Step/function is user-defined eg word 1 → count 1
word 2 → count 2

→ Next Step, is Shuffle by Key/Word
Example:- Word 'The' occurence from every worker in map stage is taken and grouped into a single worker from

Reduce stage

→ Reduce stage will have subsets of the total words/keys across each of its workers. But what it essentially does is it Aggregates the count of each word across diff Map workers and

- For Map-Reduce we do not have to worry about the underlying infrastructure as in the no of workers we need so on
  Example:- Batch Processing nowadays is done using Spark/Hadoop cluster

NOTE: Batch Processing Example -
Spark cluster also has micro batching substitute for streaming
Streaming Example -
Apache Flink